

Supplemental Material

Table S1. SNPs of serum uric acid used to construct the instrumental variable for the MR analysis.

	SNP	β	SE	<i>P</i> value	R2	F
1	rs926979	0.0137	0.002	3.35E-12	7.63E-05	46.9225
2	rs12096443	-0.0174	0.0019	1.54E-19	0.000137	83.86704
3	rs11163481	0.0125	0.0018	1.24E-11	7.17E-05	48.22531
4	rs2493121	0.0121	0.002	1.05E-09	6.01E-05	36.6025
5	rs2808500	-0.0104	0.0018	5.89E-09	5.31E-05	33.38272
6	rs2644128	0.0136	0.0018	7.44E-14	9.22E-05	57.08642
7	rs144188351	0.0111	0.002	1.60E-08	5.68E-05	30.8025
8	rs6674490	-0.0123	0.0018	9.68E-12	7.22E-05	46.69444
9	rs10922199	0.0106	0.0019	1.84E-08	5.28E-05	31.12465
10	rs12070208	-0.0211	0.0038	3.16E-08	6.61E-05	30.83172
11	rs496708	-0.0133	0.0018	7.26E-13	8.75E-05	54.59568
12	rs11264341	-0.0352	0.0018	1.57E-82	0.000619	382.4198
13	rs6429747	-0.0215	0.0022	9.60E-23	0.000199	95.5062
14	rs11164916	0.0188	0.0018	1.79E-25	0.000169	109.0864
15	rs17024258	0.036	0.0064	2.12E-08	6.53E-05	31.64063
16	rs199661312	0.039	0.0071	4.25E-08	6.24E-05	30.17258
17	rs150147865	-0.1163	0.0116	8.70E-24	0.000238	100.5179
18	rs140885868	-0.0893	0.0097	2.40E-20	0.00021	84.75385
19	rs3845534	0.0151	0.0018	3.57E-16	0.000111	70.37346
20	rs2480714	0.0126	0.0022	5.87E-09	5.69E-05	32.80165
21	rs114165349	0.0805	0.0067	8.01E-33	0.000298	144.3584
22	rs2788144	0.0395	0.0054	3.38E-13	0.00011	53.50652
23	rs13418518	0.0366	0.0058	2.22E-10	8.35E-05	39.82045
24	rs1260326	-0.0445	0.0018	1.88E-136	0.000975	611.1883
25	rs10196697	-0.0105	0.0019	2.71E-08	5.51E-05	30.54017
26	rs10211562	-0.0134	0.0018	4.31E-14	8.88E-05	55.41975
27	rs12467636	0.013	0.0021	3.59E-10	6.24E-05	38.322
28	rs2287084	0.0124	0.0021	6.36E-09	5.41E-05	34.86621
29	rs2252862	-0.0147	0.0019	6.01E-15	0.000108	59.85873
30	rs2075252	-0.0261	0.002	1.69E-40	0.0003	170.3025
31	rs187355703	0.0563	0.0064	1.32E-18	0.000119	77.38501
32	rs12992672	0.0156	0.0024	2.10E-10	6.25E-05	42.25
33	rs17050272	0.0218	0.0018	1.59E-34	0.000233	146.679
34	rs13021972	0.016	0.0018	4.50E-18	0.000124	79.01235
35	rs146787580	0.0264	0.0032	2.51E-16	0.000141	68.0625
36	rs1851285	0.013	0.0018	1.34E-12	8.36E-05	52.16049
37	rs807624	-0.0168	0.0019	8.23E-19	0.000141	78.18283
38	rs12105304	0.0206	0.0021	7.91E-22	0.000153	96.22676
39	rs72818964	0.0155	0.0027	5.59E-09	6.97E-05	32.9561
40	rs1021409	0.0188	0.0024	4.18E-15	9.65E-05	61.36111

41	rs13411042	0.0145	0.0018	4.33E-16	0.000104	64.89198
42	rs2012385	-0.0181	0.0021	2.72E-17	0.000148	74.28798
43	rs62106258	-0.0342	0.0047	5.11E-13	0.000108	52.94885
44	rs2060824	0.0135	0.0018	3.28E-14	9.10E-05	56.25
45	rs12987661	-0.0302	0.0029	4.22E-25	0.000165	108.4471
46	rs6715785	-0.0116	0.0021	3.98E-08	6.54E-05	30.51247
47	rs7586329	0.0641	0.0116	3.25E-08	6.32E-05	30.53515
48	rs831036	0.0146	0.0018	7.48E-16	0.000102	65.79012
49	rs1047891	-0.0271	0.002	1.54E-41	0.00029	183.6025
50	rs2943645	0.0134	0.002	2.96E-11	7.25E-05	44.89
51	rs11128603	-0.0203	0.003	9.38E-12	7.19E-05	45.78778
52	rs6804946	-0.0216	0.0025	1.77E-17	0.000149	74.6496
53	rs200216446	0.0231	0.0033	4.39E-12	7.45E-05	49
54	rs7642977	0.0106	0.0019	3.94E-08	5.46E-05	31.12465
55	rs62262727	0.0358	0.004	3.53E-19	0.000166	80.1025
56	rs146712725	-0.0139	0.0021	3.51E-11	7.64E-05	43.81179
57	rs9872445	-0.0126	0.002	8.31E-10	7.82E-05	39.69
58	rs1511299	-0.0211	0.002	4.15E-26	0.000174	111.3025
59	rs9843304	0.0103	0.0018	5.07E-09	5.28E-05	32.74383
60	rs7039	0.0147	0.0018	2.49E-16	0.000107	66.69444
61	rs7651369	-0.011	0.0019	1.31E-08	6.04E-05	33.51801
62	rs55756439	0.0177	0.0029	7.30E-10	7.94E-05	37.25208
63	rs9880232	-0.0144	0.0022	3.93E-11	7.24E-05	42.84298
64	rs79141906	-0.0123	0.0022	1.58E-08	6.64E-05	31.25826
65	rs2581824	0.03	0.0018	4.94E-64	0.000445	277.7778
66	rs62294340	-0.017	0.0018	9.99E-21	0.000135	89.19753
67	rs73169739	0.0242	0.0036	2.90E-11	6.84E-05	45.18827
68	rs6825697	0.0365	0.0038	1.12E-21	0.000201	92.26108
69	rs62409885	0.0615	0.0023	5.35E-164	0.001747	714.9811
70	rs62294282	0.3013	0.0022	1.00E-200	0.036287	18756.55
71	rs9307594	0.0105	0.0019	2.92E-08	5.39E-05	30.54017
72	rs200419650	-0.0164	0.0019	9.82E-18	0.000123	74.50416
73	rs114158982	0.0584	0.0077	4.94E-14	0.000117	57.52336
74	rs45499402	0.1462	0.0024	1.00E-200	0.005854	3710.84
75	rs12510175	-0.0117	0.0019	3.45E-10	6.69E-05	37.91967
76	rs5856715	0.0181	0.0029	6.20E-10	8.03E-05	38.95482
77	rs7677783	-0.0114	0.0018	3.34E-10	6.18E-05	40.11111
78	rs145157727	-0.0518	0.0082	2.98E-10	6.08E-05	39.90541
79	rs2231145	-0.0787	0.0046	3.35E-64	0.000506	292.7075
80	rs7696556	-0.0168	0.0023	2.48E-13	0.000112	53.3535
81	rs112214565	0.0578	0.0083	4.34E-12	0.000114	48.49528
82	rs35523212	-0.0448	0.0018	7.64E-130	0.001002	619.4568

83	rs34811474	-0.014	0.0024	6.04E-09	6.99E-05	34.02778
84	rs11940694	0.0106	0.0018	3.68E-09	5.55E-05	34.67901
85	rs17013743	-0.0298	0.0034	8.66E-19	0.00013	76.82007
86	rs142929199	-0.0527	0.0074	8.06E-13	0.000127	50.71749
87	rs13107325	-0.031	0.0039	9.97E-16	0.000133	63.18212
88	rs157512	-0.0202	0.0025	7.68E-16	0.000148	65.2864
89	rs272889	0.013	0.0018	1.85E-12	8.38E-05	52.16049
90	rs140753856	-0.0147	0.0025	6.14E-09	5.52E-05	34.5744
91	rs255749	0.0159	0.002	1.50E-15	9.96E-05	63.2025
92	rs6860575	-0.0114	0.0018	1.88E-10	6.48E-05	40.11111
93	rs2973444	-0.0245	0.0036	1.07E-11	7.16E-05	46.31559
94	rs28419182	0.018	0.002	2.87E-19	0.000138	81
95	rs17632159	-0.0299	0.0019	3.10E-54	0.000375	247.6482
96	rs455213	0.0165	0.0018	1.44E-19	0.000131	84.02778
97	rs10075612	0.0139	0.0023	1.04E-09	7.74E-05	36.52363
98	rs152197	-0.0126	0.0022	7.23E-09	5.38E-05	32.80165
99	rs11243143	-0.0353	0.0021	5.13E-63	0.000578	282.5601
100	rs833805	0.0226	0.0028	1.47E-15	0.000106	65.14796
101	rs2762353	0.0668	0.0019	1.00E-200	0.00205	1236.078
102	rs3734537	0.0401	0.003	5.29E-41	0.000371	178.6678
103	rs1317983	0.0304	0.002	2.65E-50	0.000352	231.04
104	rs6939175	-0.012	0.002	3.60E-09	5.89E-05	36
105	rs4897160	0.0145	0.0018	1.79E-16	0.000105	64.89198
106	rs2636590	0.011	0.0018	2.29E-09	5.63E-05	37.34568
107	rs676015	-0.015	0.0019	1.88E-15	0.0001	62.32687
108	rs2714345	-0.0122	0.0021	1.39E-08	5.24E-05	33.75057
109	rs148688688	0.0426	0.0065	4.42E-11	7.61E-05	42.9529
110	rs7773175	-0.0216	0.002	6.52E-28	0.000185	116.64
111	rs28755272	-0.0174	0.002	6.34E-18	0.000134	75.69
112	rs2749005	0.0116	0.002	4.67E-09	5.55E-05	33.64
113	rs34555420	0.0531	0.0034	1.40E-55	0.000511	243.9109
114	rs7766720	0.021	0.0029	6.35E-13	8.37E-05	52.43757
115	rs4476815	0.0366	0.0058	2.29E-10	6.83E-05	39.82045
116	rs72951456	0.0264	0.0048	3.17E-08	6.49E-05	30.25
117	rs505870	-0.013	0.002	1.78E-10	8.43E-05	42.25
118	rs59864860	-0.0169	0.0027	4.66E-10	6.02E-05	39.17833
119	rs856560	0.0133	0.002	2.74E-11	7.37E-05	44.2225
120	rs9333592	-0.0239	0.0039	5.83E-10	8.22E-05	37.5549
121	rs2240390	-0.0132	0.0019	7.32E-12	7.65E-05	48.26593
122	rs35994853	0.0151	0.0019	1.44E-15	0.000106	63.16066
123	rs10279504	-0.0122	0.002	9.34E-10	5.97E-05	37.21
124	rs13247874	-0.0345	0.0023	1.99E-49	0.000339	225

125	rs13221758	0.0151	0.002	9.53E-14	0.000114	57.0025
126	rs6464165	0.0267	0.0023	3.79E-32	0.000289	134.7618
127	rs13230509	0.0314	0.0019	2.61E-58	0.000477	273.1191
128	rs12666237	0.0202	0.003	3.04E-11	7.39E-05	45.33778
129	rs60767324	-0.0189	0.0033	1.20E-08	5.04E-05	32.80165
130	rs2695580	0.0107	0.0019	1.33E-08	5.34E-05	31.71468
131	rs17786744	0.0223	0.0018	1.16E-33	0.000231	153.4846
132	rs12543287	-0.0115	0.0018	3.85E-10	6.13E-05	40.8179
133	rs10429294	0.0205	0.0018	4.10E-30	0.000205	129.7068
134	rs2436958	-0.0188	0.0024	1.44E-15	0.0001	61.36111
135	rs150386926	0.0159	0.0019	2.25E-17	0.000119	70.03047
136	rs2926590	0.0193	0.0023	5.10E-17	0.000111	70.41399
137	rs1869581	-0.0109	0.0018	2.00E-09	5.87E-05	36.66975
138	rs2941484	0.0318	0.0018	1.30E-71	0.000499	312.1111
139	rs2039424	0.0111	0.0018	1.65E-09	5.69E-05	38.02778
140	rs10868758	0.0164	0.0028	2.72E-09	5.48E-05	34.30612
141	rs62580766	-0.0144	0.0026	4.75E-08	6.17E-05	30.67456
142	rs2437817	-0.0146	0.0019	2.60E-14	9.11E-05	59.04709
143	rs7341879	0.0121	0.0022	3.96E-08	6.19E-05	30.25
144	rs28517717	-0.0218	0.0019	9.78E-30	0.000213	131.6454
145	rs754600	0.0152	0.0021	1.31E-12	9.63E-05	52.39002
146	rs10901057	0.0172	0.0027	2.36E-10	7.69E-05	40.58162
147	rs56379622	-0.0374	0.0049	2.90E-14	0.000119	58.25739
148	rs3824359	0.0142	0.0023	1.53E-09	5.84E-05	38.1172
149	rs181673	0.0173	0.0018	1.56E-21	0.000145	92.37346
150	rs10886117	0.021	0.0022	5.02E-22	0.000155	91.1157
151	rs45487598	0.0184	0.0032	7.70E-09	6.99E-05	33.0625
152	rs17592117	0.0426	0.0031	1.80E-43	0.000302	188.8408
153	rs10082474	-0.0131	0.0018	1.06E-13	8.57E-05	52.96605
154	rs12775853	-0.0146	0.0022	2.11E-11	9.40E-05	44.04132
155	rs11202328	-0.0254	0.0023	2.11E-27	0.000256	121.9584
156	rs2362515	-0.0121	0.0019	2.30E-10	6.32E-05	40.55679
157	rs12767261	0.0166	0.0029	6.67E-09	5.25E-05	32.76576
158	rs7093087	0.0172	0.0024	4.35E-13	8.09E-05	51.36111
159	rs1925258	-0.0124	0.0019	5.43E-11	6.85E-05	42.5928
160	rs1171614	0.0563	0.0024	1.73E-120	0.001127	550.2934
161	rs75523587	0.0174	0.0022	1.94E-15	9.93E-05	62.55372
162	rs703978	0.0117	0.0018	1.11E-10	6.48E-05	42.25
163	rs12806743	-0.0138	0.0021	9.56E-11	6.52E-05	43.18367
164	rs3116068	0.0184	0.0022	1.67E-17	0.000113	69.95041
165	rs61903695	0.012	0.0021	2.13E-08	4.89E-05	32.65306
166	rs680071	0.0158	0.0026	1.50E-09	5.66E-05	36.92899

167	rs12277177	0.0157	0.0027	4.12E-09	7.13E-05	33.81207
168	rs7122026	0.0121	0.002	1.39E-09	6.78E-05	36.6025
169	rs589852	-0.0288	0.0018	3.84E-58	0.000399	256
170	rs7952306	0.025	0.0024	6.14E-26	0.000214	108.5069
171	rs1622987	0.011	0.002	4.54E-08	4.83E-05	30.25
172	rs2195525	-0.014	0.0019	6.72E-14	9.40E-05	54.29363
173	rs12790943	-0.014	0.0021	8.71E-12	8.33E-05	44.44444
174	rs11827106	-0.0124	0.0022	3.33E-08	5.25E-05	31.7686
175	rs11564722	-0.0204	0.002	1.45E-24	0.000174	104.04
176	rs963837	-0.0245	0.0018	4.72E-43	0.000293	185.2623
177	rs12575710	0.0264	0.0024	4.16E-29	0.00023	121
178	rs538737	0.0786	0.002	1.00E-200	0.003035	1544.49
179	rs34381009	-0.0387	0.0023	2.10E-61	0.000421	283.1172
180	rs10899125	0.0226	0.0041	4.17E-08	6.75E-05	30.3843
181	rs17473411	0.0174	0.0027	7.55E-11	8.83E-05	41.53086
182	rs836968	-0.014	0.0019	3.59E-13	8.87E-05	54.29363
183	rs11109717	-0.0137	0.002	2.18E-11	7.17E-05	46.9225
184	rs10859925	0.0159	0.0023	1.21E-11	7.26E-05	47.79017
185	rs2229357	-0.0449	0.0022	1.12E-90	0.000643	416.531
186	rs7310615	-0.0289	0.002	3.63E-45	0.000417	208.8025
187	rs74397112	0.0227	0.0029	5.19E-15	9.58E-05	61.27111
188	rs11056396	-0.0192	0.0029	3.84E-11	6.78E-05	43.83353
189	rs2240193	-0.0258	0.0037	2.09E-12	0.000305	48.62235
190	rs76895963	-0.0557	0.0078	9.75E-13	0.000128	50.99425
191	rs11056306	-0.0108	0.0018	1.17E-09	5.72E-05	36
192	rs148179165	0.0164	0.0019	3.20E-18	0.000122	74.50416
193	rs11065983	-0.0313	0.0034	3.29E-20	0.000281	84.74827
194	rs61941404	-0.028	0.0037	2.41E-14	0.000104	57.26808
195	rs1169288	-0.0202	0.0019	1.83E-27	0.000189	113.0305
196	rs10771025	-0.0118	0.002	7.17E-09	5.35E-05	34.81
197	rs7302925	-0.0184	0.0023	2.30E-15	9.68E-05	64
198	rs9534949	-0.0133	0.0019	6.51E-12	7.31E-05	49
199	rs928122	-0.0146	0.0023	2.85E-10	8.25E-05	40.2949
200	rs7331398	-0.0148	0.0019	1.37E-14	0.00011	60.6759
201	rs77008184	0.0188	0.0029	1.70E-10	6.38E-05	42.02616
202	rs12891886	0.013	0.0021	3.23E-10	6.17E-05	38.322
203	rs2210315	0.0181	0.0027	2.27E-11	9.30E-05	44.93964
204	rs12887732	0.0114	0.002	8.45E-09	5.18E-05	32.49
205	rs73349296	-0.0146	0.0022	4.49E-11	9.01E-05	44.04132
206	rs1955949	-0.0137	0.002	2.39E-11	6.99E-05	46.9225
207	rs72681698	-0.0717	0.0097	1.63E-13	0.000112	54.63801
208	rs4777466	-0.0149	0.0019	1.84E-14	0.000102	61.49861

209	rs8039645	0.0127	0.0022	1.01E-08	5.28E-05	33.32438
210	rs7402939	0.012	0.0018	3.52E-11	6.98E-05	44.44444
211	rs34868798	-0.0141	0.0024	7.87E-09	6.83E-05	34.51563
212	rs4886755	0.0285	0.0018	2.84E-57	0.000406	250.6944
213	rs12911430	0.0123	0.002	1.82E-09	6.55E-05	37.8225
214	rs4966019	-0.0323	0.0018	1.10E-70	0.000501	322.0031
215	rs6492910	-0.0133	0.002	2.02E-11	8.14E-05	44.2225
216	rs2414064	-0.0144	0.0025	7.10E-09	6.97E-05	33.1776
217	rs12708477	-0.0124	0.0022	1.98E-08	4.88E-05	31.7686
218	rs4783505	0.0139	0.0019	6.18E-13	9.60E-05	53.52078
219	rs74606487	-0.0148	0.0027	3.95E-08	5.10E-05	30.04664
220	rs17624477	0.0252	0.0038	5.10E-11	6.62E-05	43.97784
221	rs2219647	0.0163	0.002	1.25E-15	0.000101	66.4225
222	rs4575545	-0.0267	0.0019	2.23E-43	0.000298	197.4765
223	rs77924615	-0.0151	0.0022	9.05E-12	7.38E-05	47.1095
224	rs9927317	0.0177	0.0021	1.92E-17	0.000149	71.04082
225	rs2540034	0.0109	0.0018	3.42E-09	5.94E-05	36.66975
226	rs7200986	-0.0135	0.0018	3.00E-13	8.65E-05	56.25
227	rs11396409	-0.028	0.0022	9.66E-36	0.000244	161.9835
228	rs7188156	0.0276	0.0024	2.52E-31	0.000214	132.25
229	rs57577420	0.0187	0.0028	4.62E-11	6.90E-05	44.60332
230	rs61339406	-0.0161	0.0024	1.89E-11	9.56E-05	45.00174
231	rs12937692	0.0131	0.002	1.49E-10	6.99E-05	42.9025
232	rs78671965	0.0271	0.0045	1.73E-09	7.70E-05	36.26716
233	rs11871152	0.0105	0.0018	4.67E-09	5.44E-05	34.02778
234	rs7224610	-0.0265	0.0019	6.51E-44	0.000314	194.5291
235	rs11657044	0.0268	0.0022	1.19E-34	0.00027	148.3967
236	rs4789278	-0.0179	0.0022	1.35E-15	0.000125	66.20041
237	rs1719985	-0.0118	0.0018	1.60E-10	6.72E-05	42.97531
238	rs9807214	0.0128	0.0021	1.32E-09	5.98E-05	37.15193
239	rs784257	0.0149	0.0026	1.34E-08	6.75E-05	32.84172
240	rs523288	0.0205	0.0024	1.45E-17	0.000151	72.96007
241	rs7230932	-0.0101	0.0018	4.09E-08	4.70E-05	31.48457
242	rs4468717	-0.0214	0.0038	1.74E-08	6.56E-05	31.71468
243	rs16942751	0.0176	0.0027	3.21E-11	7.73E-05	42.49108
244	rs56376587	0.0113	0.0018	6.41E-10	6.30E-05	39.41049
245	rs7247977	0.0176	0.0018	2.47E-22	0.000154	95.60494
246	rs2668423	-0.0115	0.002	1.09E-08	5.16E-05	33.0625
247	rs33938520	-0.0135	0.0023	9.79E-09	6.88E-05	34.4518
248	rs11667586	0.0181	0.0026	4.92E-12	7.38E-05	48.46302
249	rs429358	-0.0182	0.0025	6.05E-13	8.03E-05	52.9984
250	rs10405423	0.0216	0.002	2.13E-26	0.000188	116.64

251	rs12973608	-0.014	0.0019	1.99E-13	8.57E-05	54.29363
252	rs12979148	0.0149	0.0025	3.48E-09	5.46E-05	35.5216
253	rs112671669	0.0171	0.0029	3.64E-09	5.50E-05	34.76932
254	rs753009	0.0142	0.0021	3.63E-11	7.34E-05	45.72336
255	rs6040060	-0.0148	0.0023	1.63E-10	8.81E-05	41.40643
256	rs3212198	-0.0172	0.0018	4.80E-21	0.000148	91.30864
257	rs11481757	0.0141	0.0022	2.45E-10	6.66E-05	41.07645
258	rs2057291	-0.0146	0.0019	5.75E-15	9.37E-05	59.04709
259	rs6129786	-0.013	0.0021	3.40E-10	7.77E-05	38.322
260	rs6088559	0.0162	0.0021	1.02E-14	0.000125	59.5102
261	rs686364	0.0167	0.002	3.10E-17	0.000114	69.7225
262	rs2823139	0.0152	0.0019	2.41E-15	9.96E-05	64
263	rs219777	-0.0227	0.0023	4.19E-22	0.000192	97.40832
264	rs2834317	0.0192	0.0027	4.91E-13	8.20E-05	50.5679
265	rs738409	-0.0199	0.002	7.17E-23	0.00016	99.0025
266	rs2108093	0.0151	0.0023	1.19E-10	6.46E-05	43.10208
267	rs12158162	-0.0212	0.0037	1.20E-08	5.04E-05	32.8298
268	rs5951348	0.0118	0.0016	4.62E-14	6.64E-05	54.39063

SNPs, single-nucleotide polymorphisms; β , beta coefficient; Se, standard error; MR, Mendelian randomization.

Table S2. Characteristics of GWAS datasets enrolled in the MR study.

Items	GWAS ID	Consortium	Sample size	Population
Serum uric acid	ebi-a-GCST90018977	UKB	343,836	Europeans
Cardiovascular disease	Finngen_R9_FG_CVD	FinnGen	377,277	Europeans
Chronic kidney disease	finngen_R9_N14_CHRONKIDNEYDIS	FinnGen	372,250	Europeans
Diabetes mellitus	finngen_R9_T2D_WIDE	FinnGen	348,788	Europeans

Table S3. Baseline and Demographic characteristics of participants aged 45 and over with and without hyperuricemia in CHARLS 2015.

Variables	Normal uric acid (n= 8568)	Hyperuricemia (n = 773)	<i>P</i> value
Age (mean [SD]) *	61.24 (9.18)	62.93 (9.64)	<0.01
Sex (%)			<0.01
Men	3762 (43.9)	617 (79.8)	
Women	4806 (56.1)	156 (20.2)	
Residence (%)			<0.01
Rural	7072 (82.5)	586 (75.8)	
Urban	1496 (17.5)	187 (24.2)	
Education level (%)			<0.01
Less than lower secondary	2400 (28.0)	133 (17.2)	
Upper secondary & vocational training	5316 (62.0)	534 (62.6)	
Tertiary	852 (9.9)	106 (13.7)	
Marital status (%)			0.084
Married/Living with partner	7391 (86.3)	684 (88.5)	
Widowed/divorced/separated	1177 (13.7)	89 (11.5)	
LDL-C (mean [SD])	2.65 (0.75)	2.65 (0.82)	0.753
Triglycerides (mean [SD])	1.59 (1.01)	1.99 (1.19)	<0.01
Uric acid (mean [SD])	4.69 (1.09)	8.02 (0.95)	<0.01
eGFR (mean [SD])	90.32(15.47)	76.58(20.73)	<0.01
BMI (%)			<0.01
< 24	4737 (55.3)	317 (41.0)	
24-28	2753 (32.1)	295 (38.2)	
≥28	1078 (12.6)	161 (20.8)	
Smoking (%)	3568 (41.6)	534 (69.1)	<0.01
Hypertension (%)	2672 (32.5)	384 (51.8)	<0.01
Cardiovascular disease (%)	1697 (19.8)	181 (23.4)	0.016
Diabetes (%)	1818 (21.2)	222 (28.7)	<0.01
Chronic kidney disease (%)	1120 (13.1)	219 (28.3)	<0.01
No. of CRM conditions (%)			<0.01
0	4980 (58.1)	336 (43.5)	
1	2663 (31.1)	285 (36.9)	
2	803 (9.4)	119 (15.4)	
3	122 (1.4)	33 (4.3)	

*Data are presented as n (percentage) for categorical variables and means and standard errors for continuous variables.

Table S4. Baseline and Demographic characteristics of participants aged 45 and over with and without hyperuricemia in NHANES, 2007-2018.

Variables	Normal uric acid (Unweighted n= 15158)	Hyperuricemia (Unweighted n = 2755)	<i>P</i> value
Weighted n (%) *	100179365 (86.6)	15467025 (13.4)	
Age (mean [SD])	60.16 (10.41)	62.49 (10.89)	<0.01
Sex (%)			<0.01
Men	6913 (43.8)	1885 (69.8)	
Women	8245 (56.2)	870 (30.2)	
Race (%)			<0.01
Mexican American	2176 (6.0)	257 (4.2)	
Other Hispanic	1676 (4.5)	220 (4.1)	
Non-Hispanic White	6728 (73.7)	1199 (70.6)	
Non-Hispanic Black	2920 (8.9)	822 (14.3)	
Other race	1658 (7.0)	257 (6.9)	
Education level (%)			<0.01
Less than 9th grade	2080 (6.3)	363 (7.1)	
9-11th grade	2093 (10.0)	409 (11.2)	
High school graduate	3467 (23.5)	667 (24.9)	
College or AA degree	4073 (29.4)	778 (32.1)	
College graduate or above	3429 (30.7)	533 (24.6)	
Marital status (%)			0.817
Married/Living with partner	9226 (66.9)	1672 (66.3)	
Widowed/divorced/separated	4769 (26.6)	867 (26.8)	
Never married	1153 (6.4)	214 (6.9)	
Poverty (%)	2508 (10.6)	498 (13.2)	<0.01
LDL-C (mean [SD])	3.03 (0.93)	2.93 (1.01)	0.03
Triglycerides (mean [SD])	1.42 (1.04)	1.80 (1.46)	<0.01
Uric acid (mean [SD])	5.14 (1.06)	7.95 (0.87)	<0.01
eGFR (mean [SD])	84.46 (18.10)	71.13 (22.06)	<0.01
BMI (%)			<0.01
< 25.0	4014 (27.3)	360 (12.1)	
25.0-29.9	5328 (35.5)	889 (33.3)	
≥ 30.0	5641 (37.2)	1441 (54.6)	
Smoking (%)	7169 (47.0)	1540 (54.4)	<0.01
Hypertension (%)	7315 (43.6)	1944 (67.0)	<0.01
Gout (%)	817 (4.9)	497 (16.8)	<0.01
Cardiovascular disease (%)	2156 (11.8)	720 (22.9)	<0.01
Diabetes (%)	3542 (17.5)	885 (26.2)	<0.01
Chronic kidney disease (%)	3424 (18.4)	1326 (42.1)	<0.01
No. of CRM conditions (%)			<0.01
0	8676 (64.9)	986 (43.1)	

1	4289 (24.5)	867 (30.0)
2	1717 (8.5)	629 (19.1)
3	476 (2.1)	273 (7.8)

*Data are presented as unweighted n (weighted percentage) for categorical variables and weighted means and standard errors for continuous variables.

Table S5. Baseline and Demographic characteristics of patients with at least 1 CRM conditions aged 45 and over with and without hyperuricemia in CHARLS.

Variables	Normal uric acid (n= 3588)	Hyperuricemia (n = 437)	P value
Age (mean [SD]) *	62.87 (9.23)	64.79(9.46)	<0.01
Sex (%)			<0.01
Men	1511 (42.1)	320 (73.2)	
Women	2077 (57.9)	117(26.8)	
Education level (%)			<0.01
1	1061 (29.6)	84 (19.2)	
2	2166 (60.4)	297 (68.0)	
3	361(10.1)	56 (12.8)	
Residence (%)			<0.001
Rural	2846(79.3)	314(71.9)	
Urban	742(20.7)	123(28.1)	
Marital status (%)			0.182
Married/Living with partner	3024 (84.3)	379(86.7)	
Widowed/divorced/separated	564(15.7)	58(13.3)	
LDL-C (mean [SD])	2.67 (0.78)	2.68 (0.80)	0.751
Triglycerides (mean [SD])	1.73 (1.09)	2.06 (1.21)	<0.01
Uric acid (mean [SD])	4.76 (1.10)	8.10 (1.03)	<0.01
eGFR (mean [SD])	86.51 (18.99)	69.94 (22.99)	<0.01
BMI (%)			<0.01
<24	1744 (48.6)	161(36.8)	
24-28	1255 (35.0)	171 (39.1)	
≥28	588 (16.4)	105 (24.0)	
Smoking (%)	1478 (41.5)	278 (63.6)	<0.01
Hypertension (%)	1576 (46.0)	261 (62.1)	<0.01
Cardiovascular disease (%)	1697(47.3)	181 (41.4)	0.020
Diabetes (%)	1818 (50.7)	222 (50.8)	0.958
Chronic kidney disease (%)	1120(31.2)	219 (50.1)	<0.01
No. of CRM conditions (%)			<0.01
1	2663 (74.2)	285(65.2)	
2	803(22.4)	119 (27.2)	
3	122 (3.4)	33 (7.6)	

*Data are presented as n (percentage) for categorical variables and weighted means and standard errors for continuous variables.

Table S6. Baseline and Demographic characteristics of patients with at least 1 CRM conditions aged 45 and over with and without hyperuricemia in NHANES, 2007-2018.

Variables	Normal uric acid (Unweighted n= 6482)	Hyperuricemia (Unweighted n = 1769)	P value
Weighted n (%)	35145738 (79.9)	8802588 (20.1)	
Age (mean [SD]) *	64.79 (10.74)	66.58 (10.55)	<0.01
Sex (%)			<0.01
Men	3128 (45.8)	1102 (60.8)	
Women	3354 (54.2)	667 (39.2)	
Race (%)			<0.01
Mexican American	960 (6.9)	161 (4.1)	
Other Hispanic	653 (4.6)	130 (3.4)	
Non-Hispanic White	2843 (69.7)	805 (69.9)	
Non-Hispanic Black	1363 (11.0)	541 (15.8)	
Other race	663 (7.8)	132 (6.8)	
Education level (%)			0.594
Less than 9th grade	1102 (9.2)	276 (9.3)	
9-11th grade	1021 (12.6)	268 (11.5)	
High school graduate	1539 (25.6)	432 (26.1)	
College or AA degree	1684 (29.9)	501 (31.1)	
College graduate or above	1129 (22.6)	287 (21.9)	
Marital status (%)			0.80
Married/Living with partner	3685 (61.7)	1003 (61.1)	
Widowed/divorced/separated	2334 (32.2)	627 (32.3)	
Never married	456 (6.1)	137 (6.6)	
Poverty (%)	1242 (14.0)	355 (15.9)	0.04
LDL-C (mean [SD])	2.72 (0.95)	2.70 (1.03)	0.80
Triglycerides (mean [SD])	1.57 (1.22)	1.73 (1.09)	<0.01
Uric acid (mean [SD])	5.25 (1.05)	7.96 (0.82)	<0.01
eGFR (mean [SD])	77.11 (22.34)	61.62 (22.52)	<0.01
BMI (%)			<0.01
<18.5	83 (1.2)	1 (0.1)	
18.5-24.9	1331 (20.1)	221 (11.2)	
25-29.9	2112 (32.1)	527 (30.5)	
≥30.0	2839 (46.6)	967 (58.2)	
Smoking (%)	3288 (51.4)	1008 (55.7)	0.024
Hypertension (%)	4270 (63.9)	1401 (77.8)	<0.01
Gout (%)	567 (7.9)	371 (21.1)	<0.01
Cardiovascular disease (%)	2156 (33.7)	720 (40.2)	<0.01
Diabetes (%)	3542 (50.0)	885 (46.0)	<0.01
Chronic kidney disease (%)	3424 (52.3)	1326 (73.9)	<0.01
No. of CRM conditions (%)			<0.01

1	4289 (24.5)	867 (30.0)
2	1717 (8.5)	629 (19.1)
3	476 (2.1)	273 (7.8)

*Data are presented as unweighted n (weighted percentage) for categorical variables and weighted means and standard errors for continuous variables.

Table S7. Mortality rates of patients aged 45 and over with different CRM conditions in both samples.

	Death/Cases	Mortality rate
CHARLS*		
CRM conditions = 1	248/2948	8.4%
CVD	94/1004	9.4%
CKD	64/697	9.2%
DM	90/1247	7.2%
CRM conditions = 2	124/922	13.4%
CVD+CKD	35/284	12.3%
CVD+DM	54/435	12.4%
CKD+DM	35/203	17.2%
CRM conditions = 3	28/155	18.1%
CVD+CKD+DM	28/155	18.1%
CRM conditions $\geq$ 1	400/4025	9.9%
NHANES		
CRM conditions = 1	901/5156	17.5%
CVD	213/1009	21.1%
CKD	513/2106	24.4%
DM	175/2041	8.57%
CRM conditions = 2	701/2346	29.9%
CVD+CKD	293/667	43.9%
CVD+DM	94/451	20.8%
CKD+DM	314/1228	25.6%
CRM conditions = 3	344/749	45.9%
CVD+CKD+DM	344/707	45.9%
CRM conditions $\geq$ 1	1946/8251	23.6%

* 5-year mortality rate

Table S8. Stratified analysis of odds ratios for the relationship among different serum uric acid levels and CRM conditions in CHARLS 2015.

	CRM conditions Model 2
Aged 45-64	
SUA (mg/dL)	1.019 (0.974-1.066)
Normal uric acid	1.0 (ref)
Hyperuricemia	1.269 (1.033-1.559) *
Aged 65 and above	
SUA (mg/dL)	1.166 (1.109-1.227) *
Normal uric acid	1.0 (ref)
Hyperuricemia	1.795 (1.436-2.244) **
Males	
SUA (mg/dL)	1.017 (0.972-1.064)
Normal uric acid	1.0 (ref)
Hyperuricemia	1.271 (1.068-1.514) *
Females	
SUA (mg/dL)	1.158(1.103-1.217) **
Normal uric acid	1.0 (ref)
Hyperuricemia	2.462 (1.810-3.349) **
Normal	
SUA (mg/dL)	1.201 (1.057-1.364) **
Normal uric acid	1.0 (ref)
Hyperuricemia	2.974 (1.510-5.860) **
Overweight	
SUA (mg/dL)	1.102(1.041-1.166) **
Normal uric acid	1.0 (ref)
Hyperuricemia	1.473 (1.102-1.968) **
Obesity	
SUA (mg/dL)	1.065 (0.989-1.473)
Normal uric acid	1.0 (ref)
Hyperuricemia	1.564 (1.132-2.161) **

* $P < 0.05$, ** $P < 0.01$.

CRM conditions: Cardiac, Renal, and Metabolic conditions. SUA: Serum uric acid

Model 2: Adjusted for age, sex, residence, education, BMI, hypertension, smoking status, and triglyceride.

Table S9. Stratified analysis of weighted odds ratios for the relationship among different serum uric acid levels, gout and CRM conditions in NHANES, 2007-2018.

	CRM conditions Model 2
Aged 45-64	
SUA (mg/dL)	1.041 (1.040-1.042) **
Normal control	1.0 (ref)
Asymptomatic HUA	1.333 (1.330-1.335) **
Non-Gout	1.0 (ref)
Gout	1.659 (1.655-1.663) **
Aged 65 and above	
SUA (mg/dL)	1.302 (1.301-1.303) **
Normal control	1.0 (ref)
Asymptomatic HUA	2.144 (2.140-2.148) **
Non-Gout	1.0 (ref)
Gout	1.716 (1.712-1.720) **
Males	
SUA (mg/dL)	1.063 (1.062-1.064) **
Normal control	1.0 (ref)
Asymptomatic HUA	1.351 (1.349-1.353) **
Non-Gout	1.0 (ref)
Gout	1.925 (1.921-1.928) **
Females	
SUA (mg/dL)	1.359 (1.358-1.360) **
Normal control	1.0 (ref)
Asymptomatic HUA	3.027 (3.020-3.033) **
Non-Gout	1.0 (ref)
Gout	1.504 (1.500-1.508) **
Normal	
SUA (mg/dL)	1.260 (1.259-1.261) **
Normal control	1.0 (ref)
Asymptomatic HUA	1.920 (1.913-1.927) **
Non-Gout	1.0 (ref)
Gout	2.514 (2.504-2.5257) **
Overweight	
SUA (mg/dL)	1.183 (1.182-1.184) **
Normal control	1.0 (ref)
Asymptomatic HUA	1.673 (1.669-1.677) **
Non-Gout	1.0 (ref)
Gout	1.522 (1.518-1.526) **
Obesity	
SUA (mg/dL)	1.109 (1.108-1.110) **

Normal control	1.0 (ref)
Asymptomatic HUA	1.593 (1.590-1.596) **
Non-Gout	1.0 (ref)
Gout	1.575 (1.572-1.579) **

* $P < 0.05$, ** $P < 0.01$.

CRM conditions: Cardiac, Renal, and Metabolic conditions. SUA: Serum uric acid.

Model 2: Adjusted for age, sex, race, education, BMI, hypertension, smoking status, poverty and triglyceride.

Table S10. Stratified analysis of hazard ratios (95% CIs) for all-cause mortality according to serum uric acid and hyperuricemia among participants with different CRM conditions (CHARLS 2015).

CRM conditions	0 CRM conditions Model 2	≥1 CRM conditions Model 2
Aged 45-64		
SUA (mg/dL)	1.084 (0.907-1.295)	1.102 (0.955-1.272)
Hyperuricemia †	1.227 (0.595-2.530)	1.680 (0.929-3.039)
Aged 65 and above		
SUA (mg/dL)	1.006 (0.880-1.150)	1.086 (0.995-1.186)
Hyperuricemia	0.815 (0.442-1.502)	1.361 (0.980-1.889)
Males		
SUA (mg/dL)	1.036 (0.919-1.168)	1.035 (0.941-1.138) *
Hyperuricemia	0.891 (0.538-1.476)	1.437 (1.031-2.002) *
Females		
SUA (mg/dL)	1.010 (0.806-1.265)	1.197 (1.063-1.348) **
Hyperuricemia	1.483 (0.446-4.932)	1.409 (0.779-2.551)
Normal		
SUA (mg/dL)	1.082 (0.957-1.224)	1.098 (1.001-1.205) *
Hyperuricemia	1.014 (0.566-1.817)	1.518 (1.035-2.228) *
Overweight		
SUA (mg/dL)	0.948 (0.762-1.179)	1.118 (0.958-1.304)
Hyperuricemia	0.808 (0.349-1.871)	1.499 (0.879-2.558)
Obesity		
SUA (mg/dL)	0.696 (0.377-1.283)	1.051 (0.844-1.308)
Hyperuricemia	0.487 (0.054-4.356)	1.166 (0.543-2.501)

* $P < 0.05$, ** $P < 0.01$.

SUA: Serum uric acid; HUA: Hyperuricemia; CRM conditions: Cardiac, Renal, and Metabolic conditions.

†: Compared with participants without HUA

Model 2: Adjusted for age, sex, residence, education, BMI, hypertension, smoking status, eGFR and triglyceride.

Table S11. Stratified analysis of weighted hazard ratios (95% CIs) for all-cause mortality according to serum uric acid, hyperuricemia and gout among participants with different CRM conditions (NHANES, 2007-2018).

CRM conditions	0 CRM conditions Model 2	≥1 CRM conditions Model 2
Aged 45-64		
SUA (mg/dL)	0.990 (0.847-1.157)	1.140 (1.037-1.253) **
Asymptomatic HUA †	1.314 (0.740-2.333)	1.376 (0.991-1.911)
Gout with normal UA †	1.514 (0.466-4.912)	1.268 (0.646-2.487)
Gout with poorly-controlled HUA †	2.676 (1.034-6.929) *	1.840 (1.009-3.355) *
Aged 65 and above		
SUA (mg/dL)	0.834 (0.698-1.003)	1.026 (0.976-1.077)
Asymptomatic HUA †	0.997 (0.687-1.447)	1.243 (1.052-1.469) *
Gout with normal UA †	- †	1.090 (0.853-1.392)
Gout with poorly-controlled HUA ^a †	- †	1.455 (1.109-1.908) **
Males		
SUA (mg/dL)	0.918 (0.809-1.042)	1.082 (1.017-1.151) *
Asymptomatic HUA †	1.366 (0.897-2.081)	1.194 (0.971-1.468)
Gout with normal UA †	1.027 (0.499-2.114)	1.106 (0.842-1.404)
Gout with poorly-controlled HUA †	1.388 (0.579-3.331)	1.322 (0.967-1.789)
Females		
SUA (mg/dL)	1.020 (0.881-1.180)	1.068 (0.985-1.158)
Asymptomatic HUA †	1.191 (0.654-2.170)	1.446 (1.145-1.827) **
Gout with normal UA †	- †	1.063 (0.678-1.668)
Gout with poorly-controlled HUA †	- †	2.237 (1.503-3.331) **
Normal		
SUA (mg/dL)	0.972 (0.851-1.112)	1.034 (0.924-1.158)
Asymptomatic HUA †	0.988 (0.435-2.241)	1.203 (0.841-1.722)
Gout with normal UA †	- †	1.351 (0.902-2.026)
Gout with poorly-controlled HUA †	- †	1.699 (0.966-2.991)
Overweight		
SUA (mg/dL)	0.958 (0.775-1.182)	1.047 (0.975-1.126)
Asymptomatic HUA †	1.644 (0.976-2.770)	1.183 (0.930-1.504)
Gout with normal UA †	0.968 (0.303-3.100)	1.177 (0.793-1.745)
Gout with poorly-controlled HUA †	1.124 (0.426-2.963)	1.645 (1.097-2.467) *
Obesity		
SUA (mg/dL)	0.975 (0.820-1.158)	1.123 (1.058-1.193) **
Asymptomatic HUA †	1.427 (0.823-2.473)	1.456 (1.182-1.793) **
Gout with normal UA †	1.170 (0.541-2.544)	1.010 (0.716-1.424)
Gout with poorly-controlled HUA †	1.541 (0.543-5.128)	1.641 (1.170-2.301) **

* $P < 0.05$, ** $P < 0.01$.

SUA: Serum uric acid; HUA: Hyperuricemia; CRM conditions: Cardiac, Renal, and Metabolic

conditions.

†: Compared with participants without HUA and/or gout.

‡: Due to the limited number of gout patients in these subgroups, only the relationship between uric acid and CRM outcome was analyzed.

Model 2: Adjusted for age, sex, race, education, poverty, BMI, hypertension, smoking status, eGFR and triglyceride.

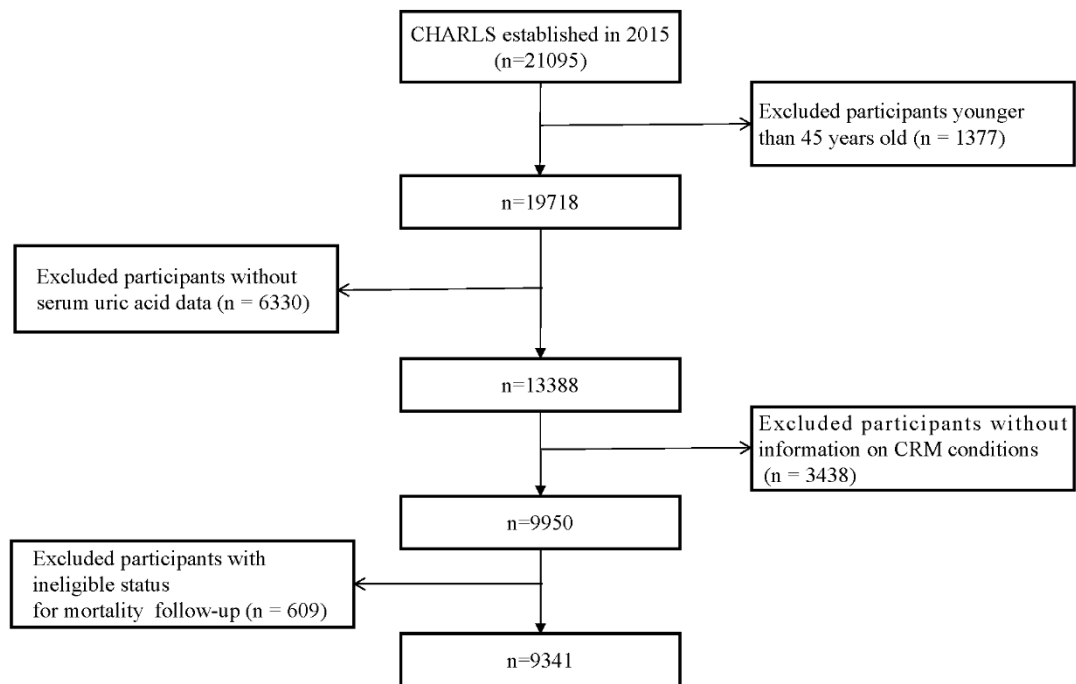


Figure S1. Flow chart of the screening process for the selection of the study population from CHARLS database. CHARLS, China Health and Retirement Longitudinal Study.

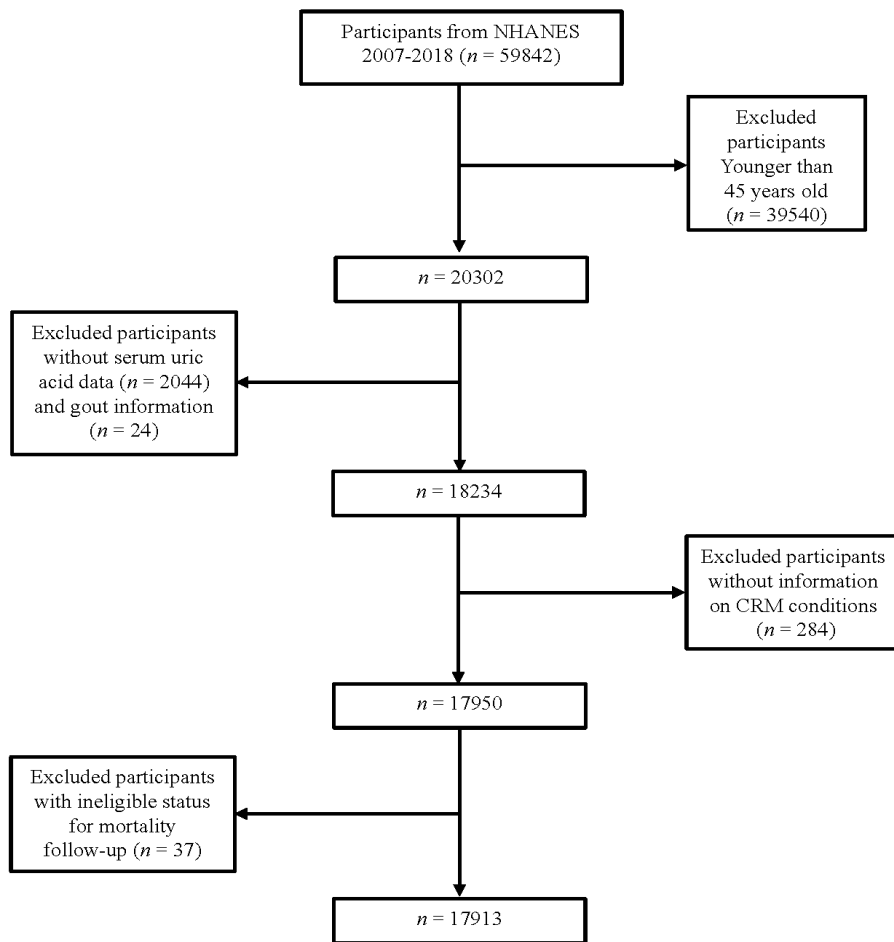


Figure S2. Flow chart of the screening process for the selection of the study population from NHANES database. NHANES, National Health and Nutrition Examination Survey.

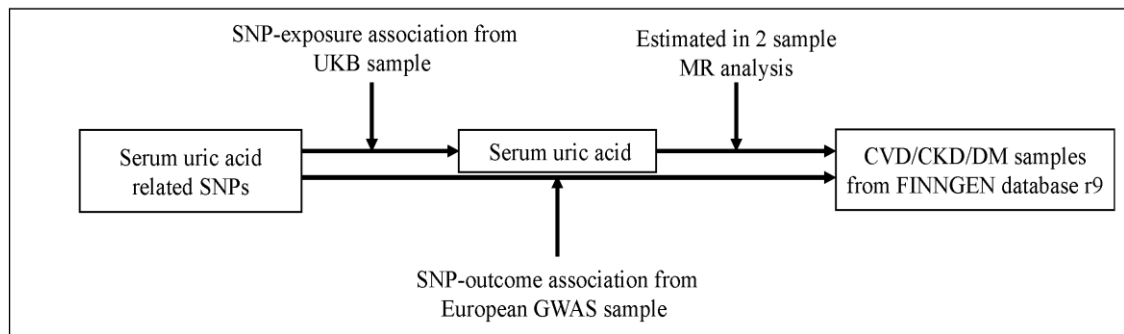


Figure S3. Two-sample Mendelian Randomization study of the association among serum uric acid with CVD, CKD and DM. CVD, cardiovascular disease; CKD, chronic kidney disease; DM, diabetes mellitus.

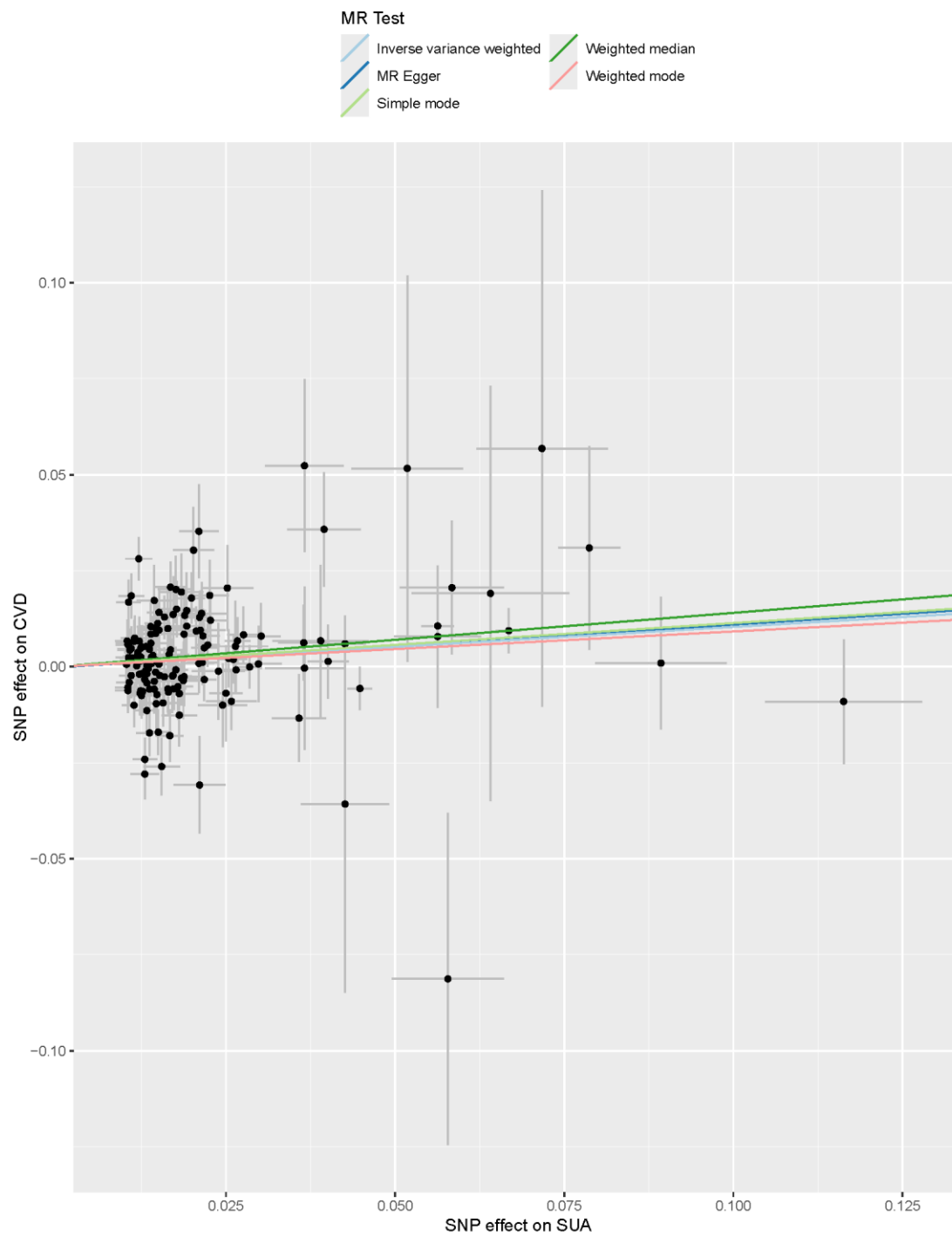


Figure S4. Scatter plot showing the associations of the SNP effects between SUA and CVD. SUA, Serum uric acid; CVD, Cardiovascular disease.

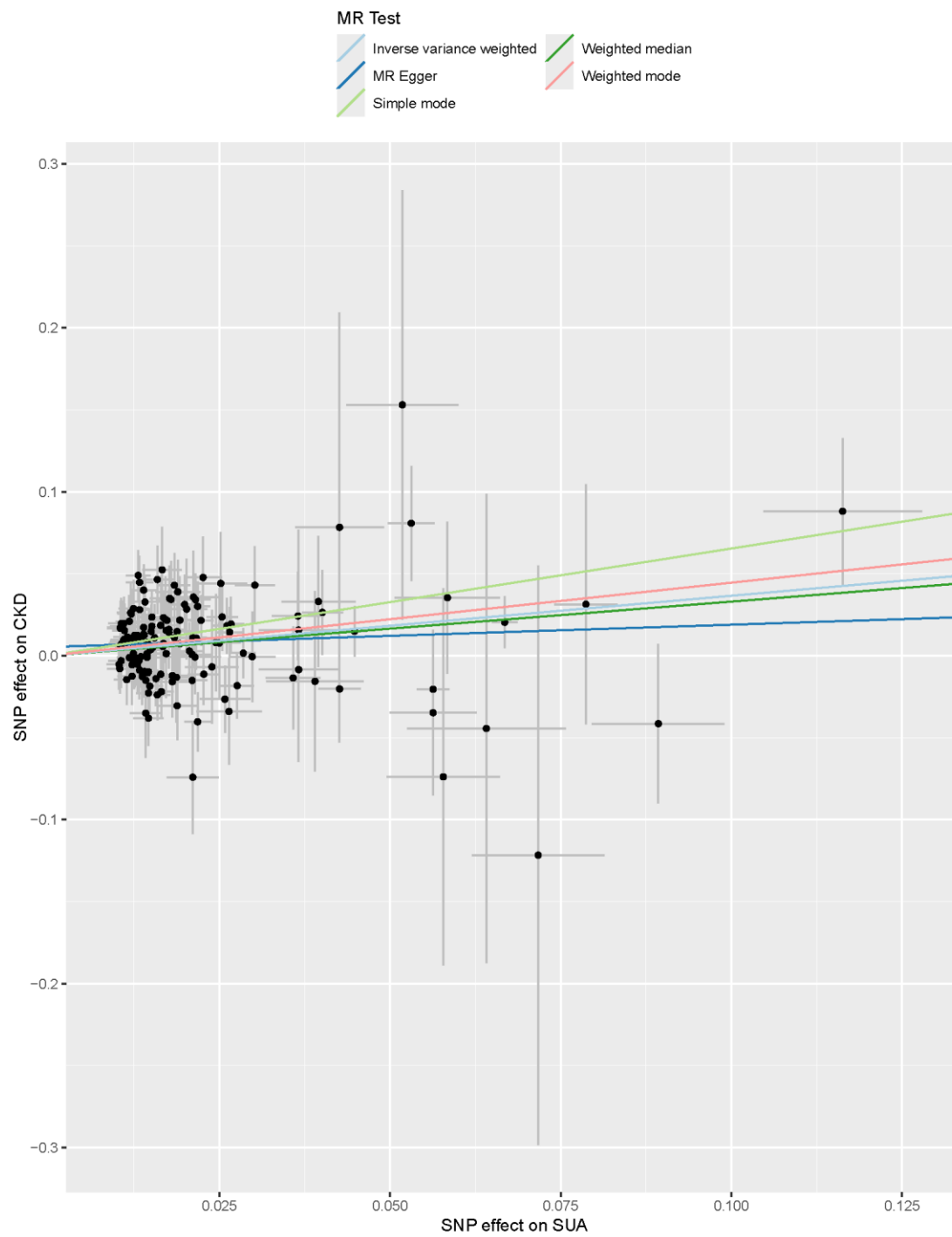


Figure S5. Scatter plot showing the associations of the SNP effects between SUA and CKD. SUA, Serum uric acid; CKD, Chronic kidney disease.

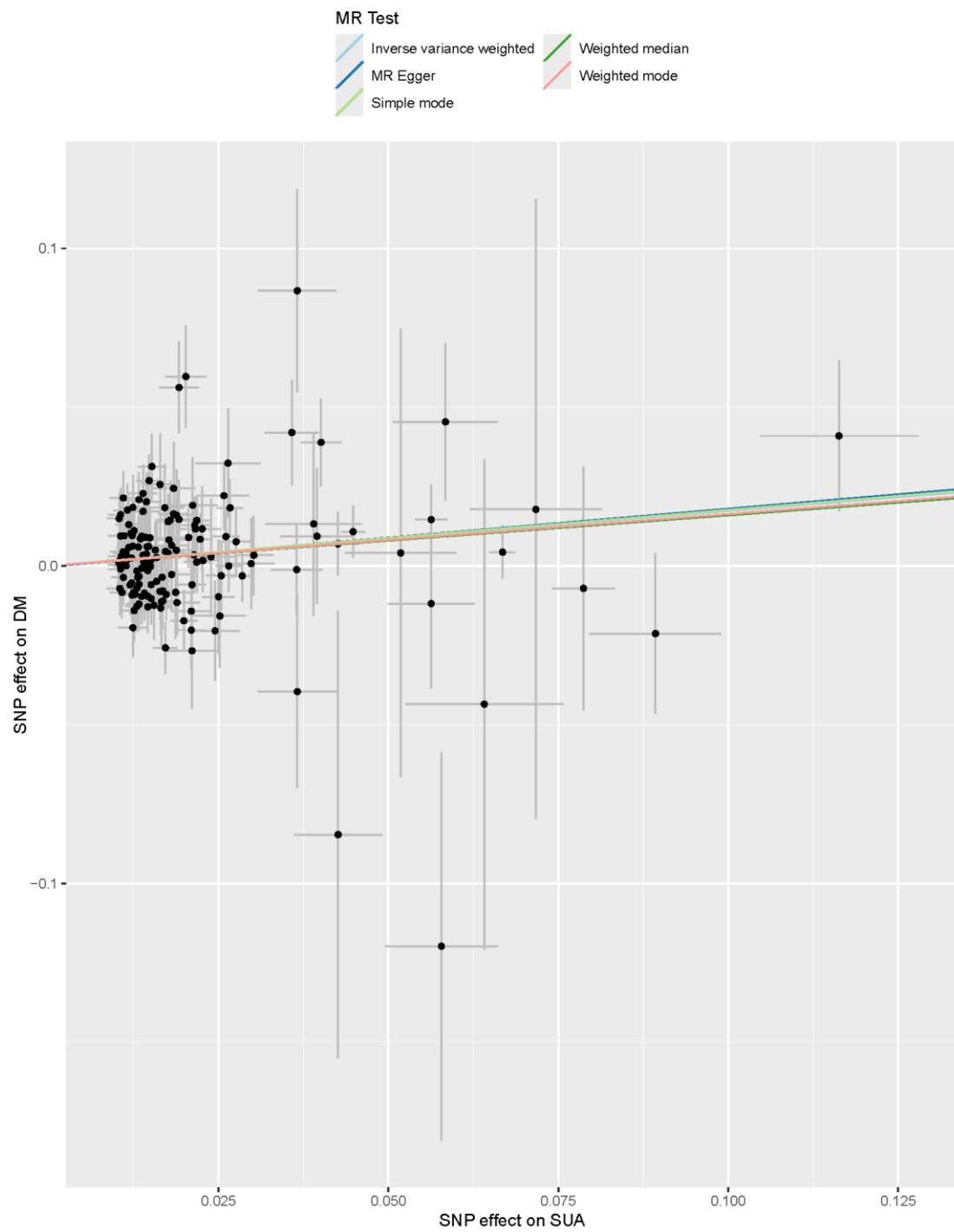


Figure S6. Scatter plot showing the associations of the SNP effects between SUA and DM. SUA, Serum uric acid; DM, Diabetes mellitus.

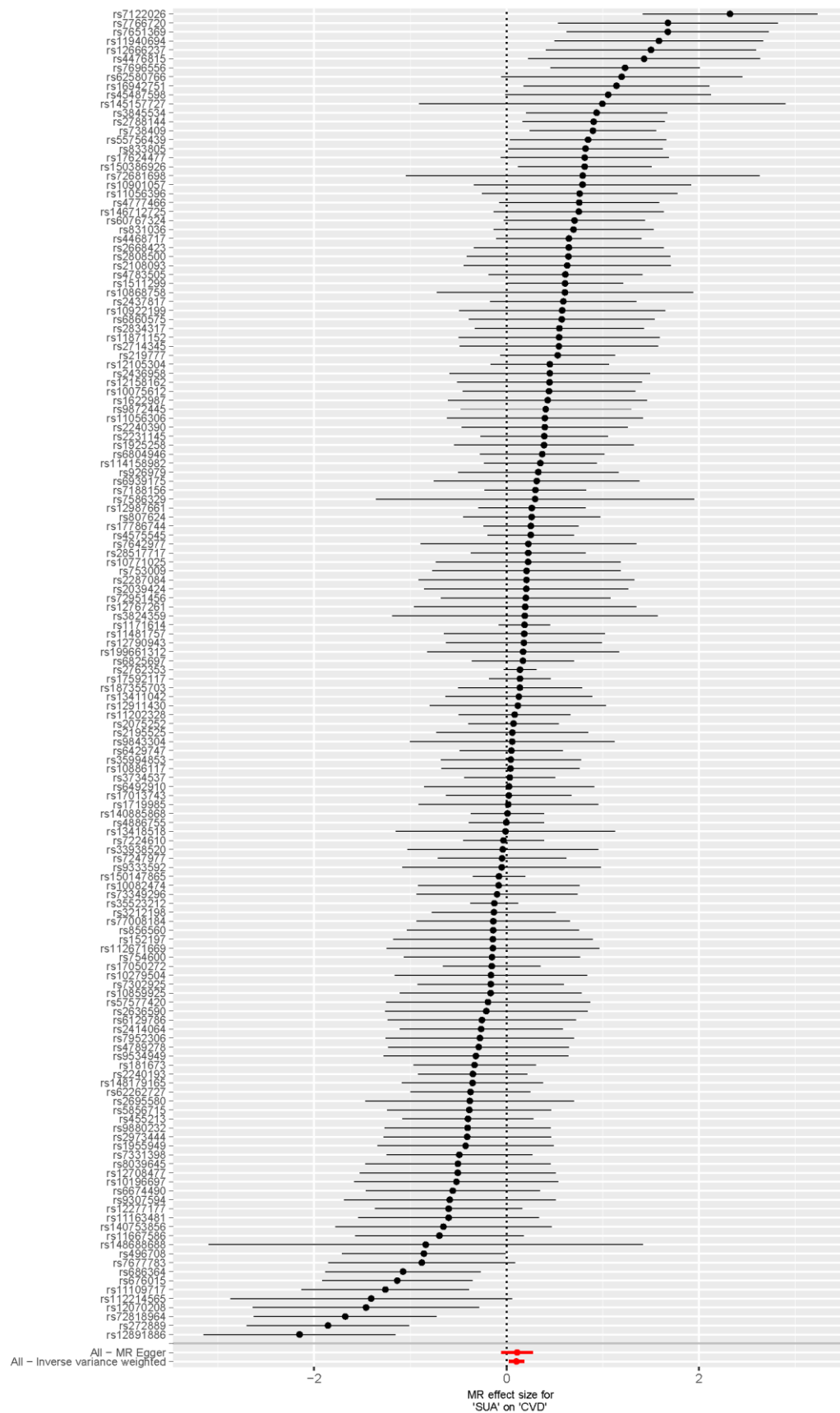


Figure S7. Forest plot of causal association for SUA on CVD. SUA, Serum uric acid; CVD, Cardiovascular disease.

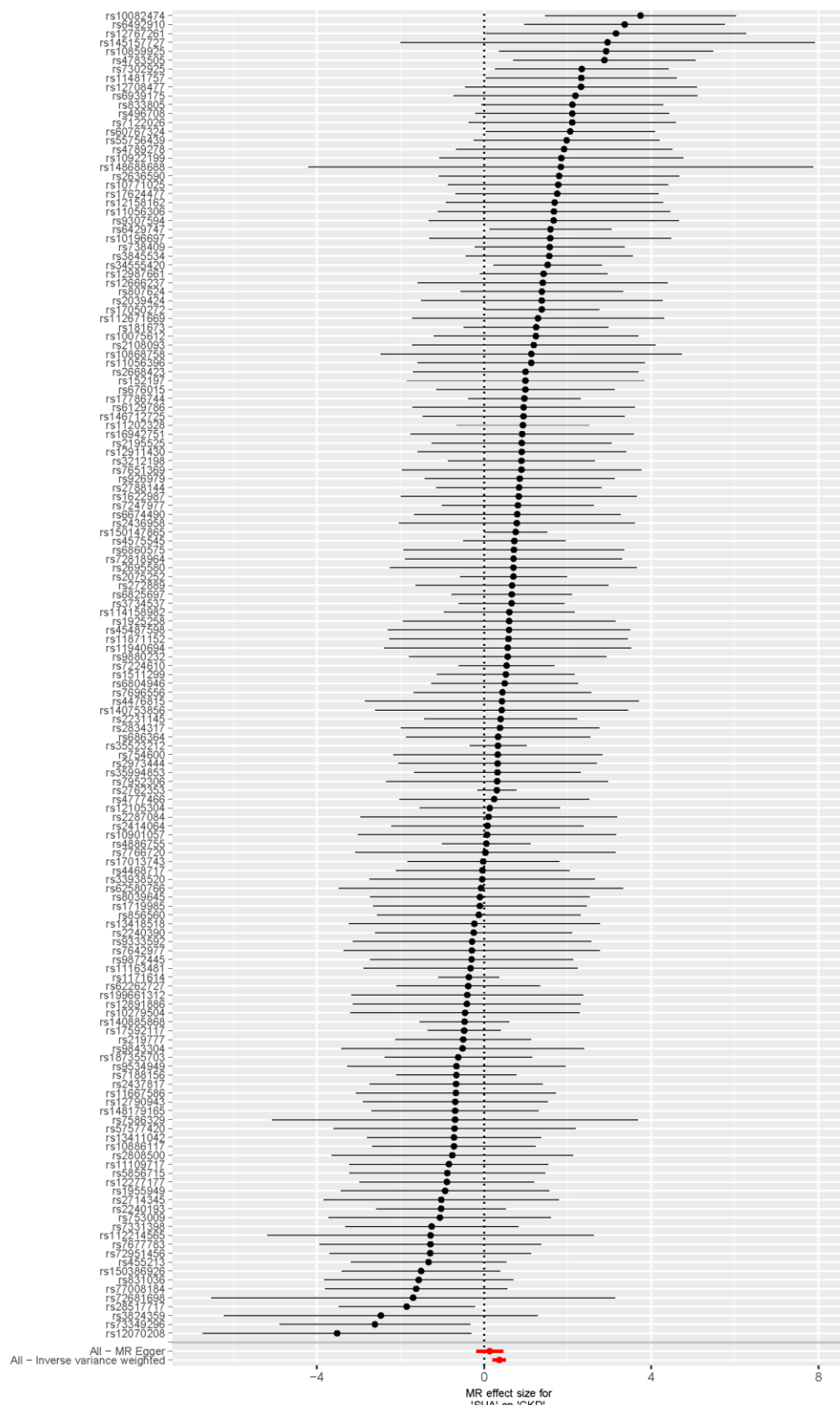


Figure S8. Forest plot of causal association for SUA on CKD. SUA, Serum uric acid; CKD, Chronic kidney disease.

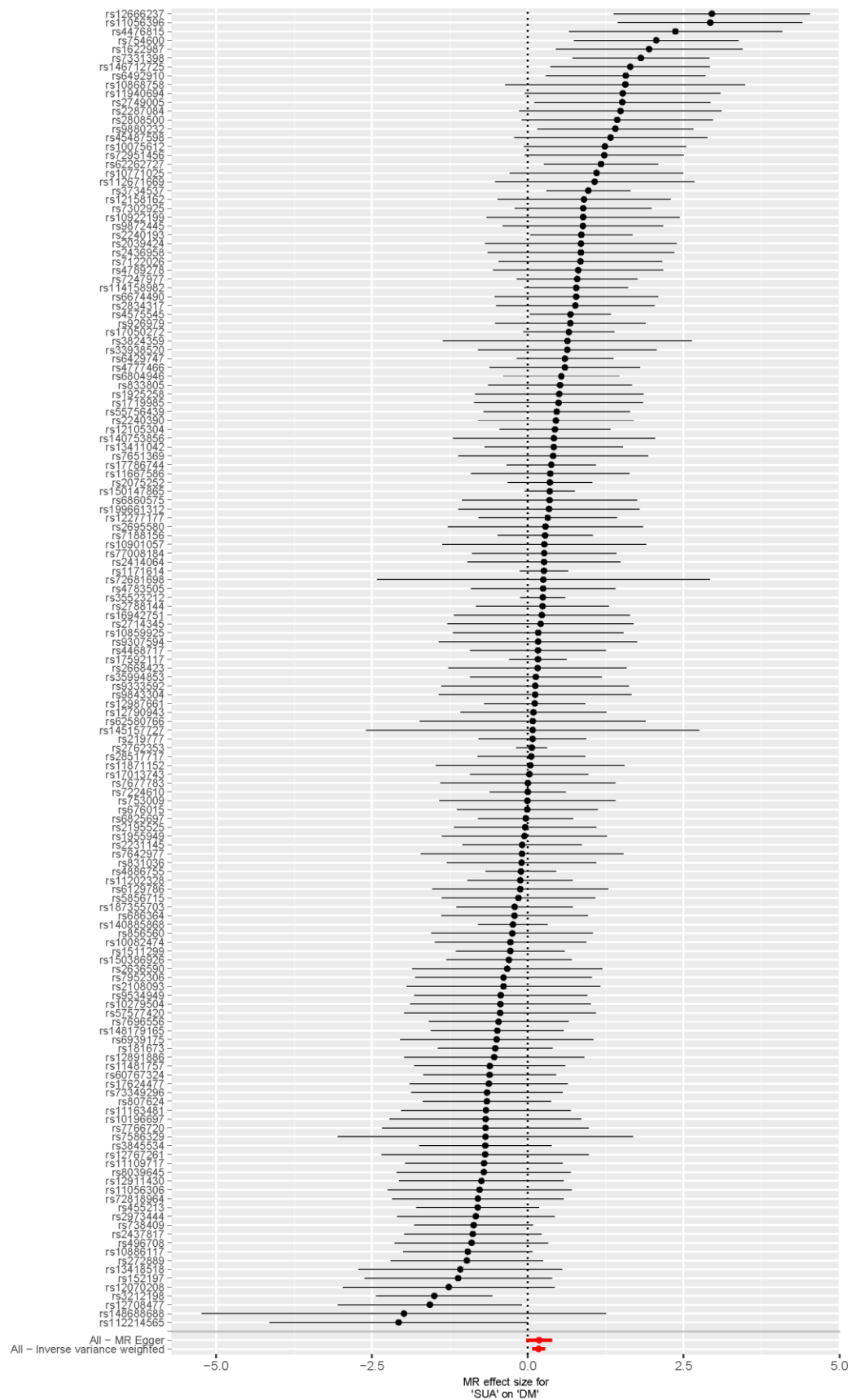


Figure S9. Forest plot of causal association for SUA on DM. SUA, Serum uric acid; DM, Diabetes mellitus.

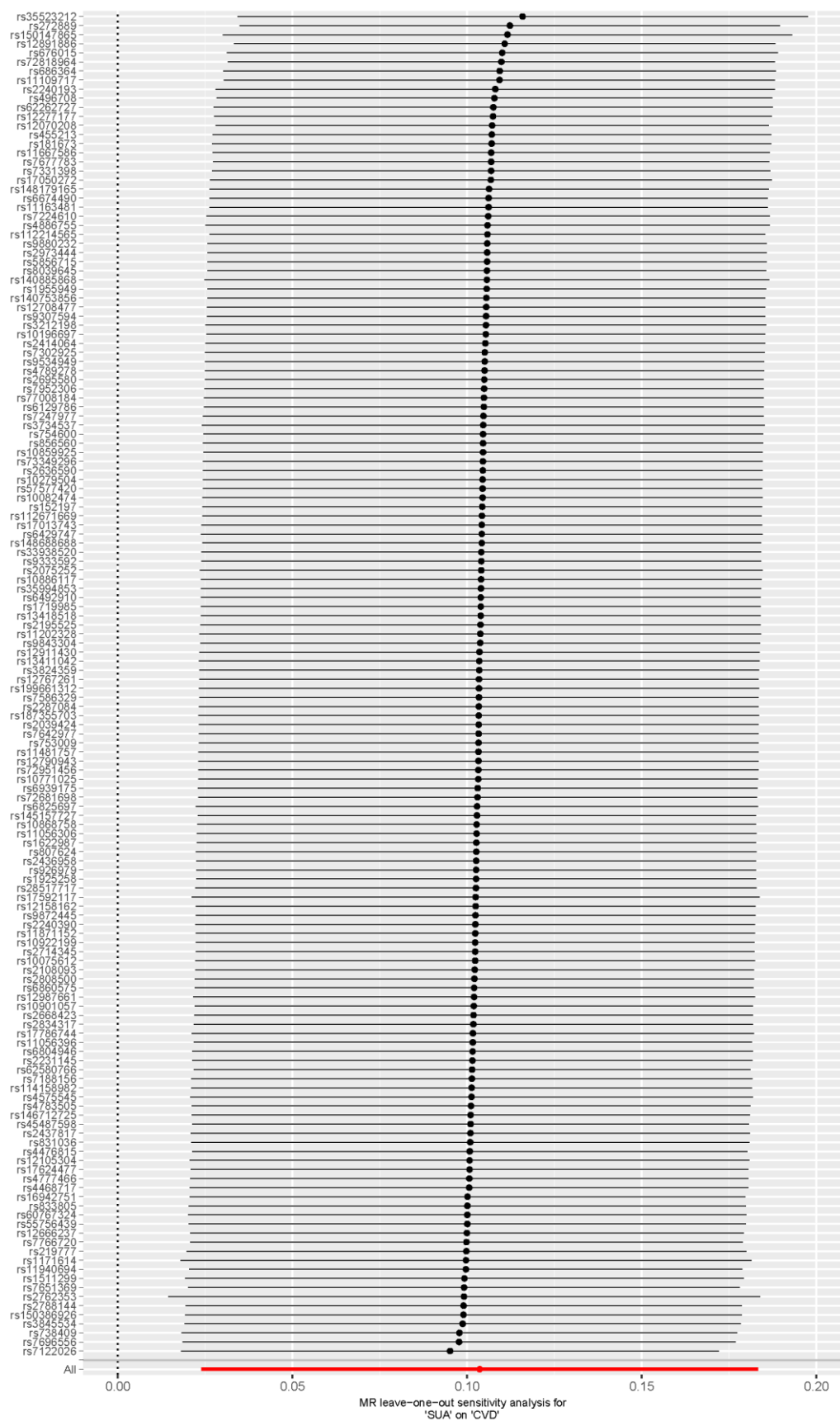


Figure S10. Leave-one-out analysis of the effect of the SUA on CVD. SUA, Serum uric acid; CVD, Cardiovascular disease.

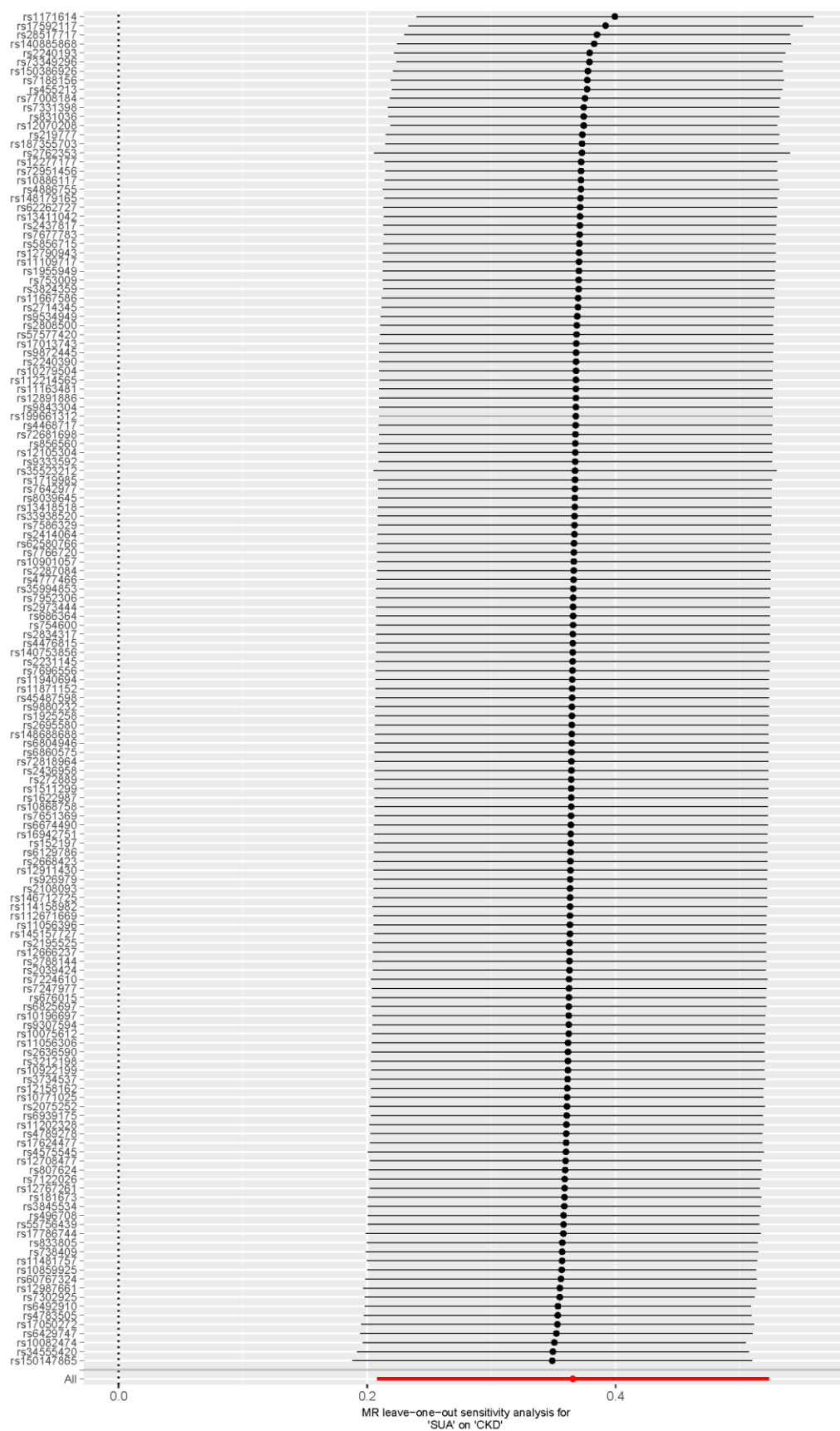


Figure S11. Leave-one-out analysis of the effect of the SUA on CKD. SUA, Serum uric acid; CKD, Chronic kidney disease.



Figure S12. Leave-one-out analysis of the effect of the SUA on DM. SUA, Serum uric acid; DM, Diabetes mellitus.

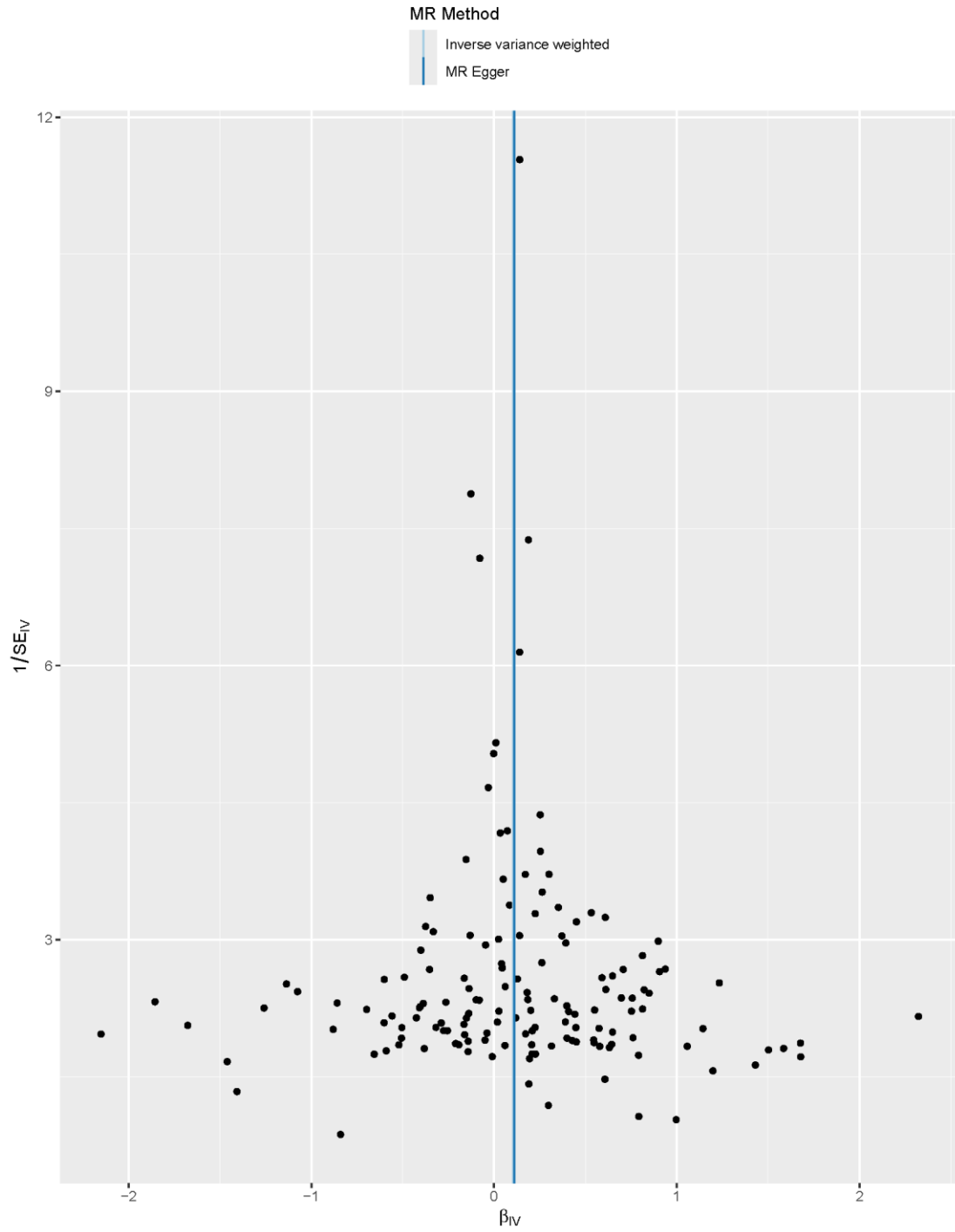


Figure S13. Funnel plots to visualize overall heterogeneity of MR estimates for the effect of serum uric acid on cardiovascular disease. MR, Mendelian Randomization.

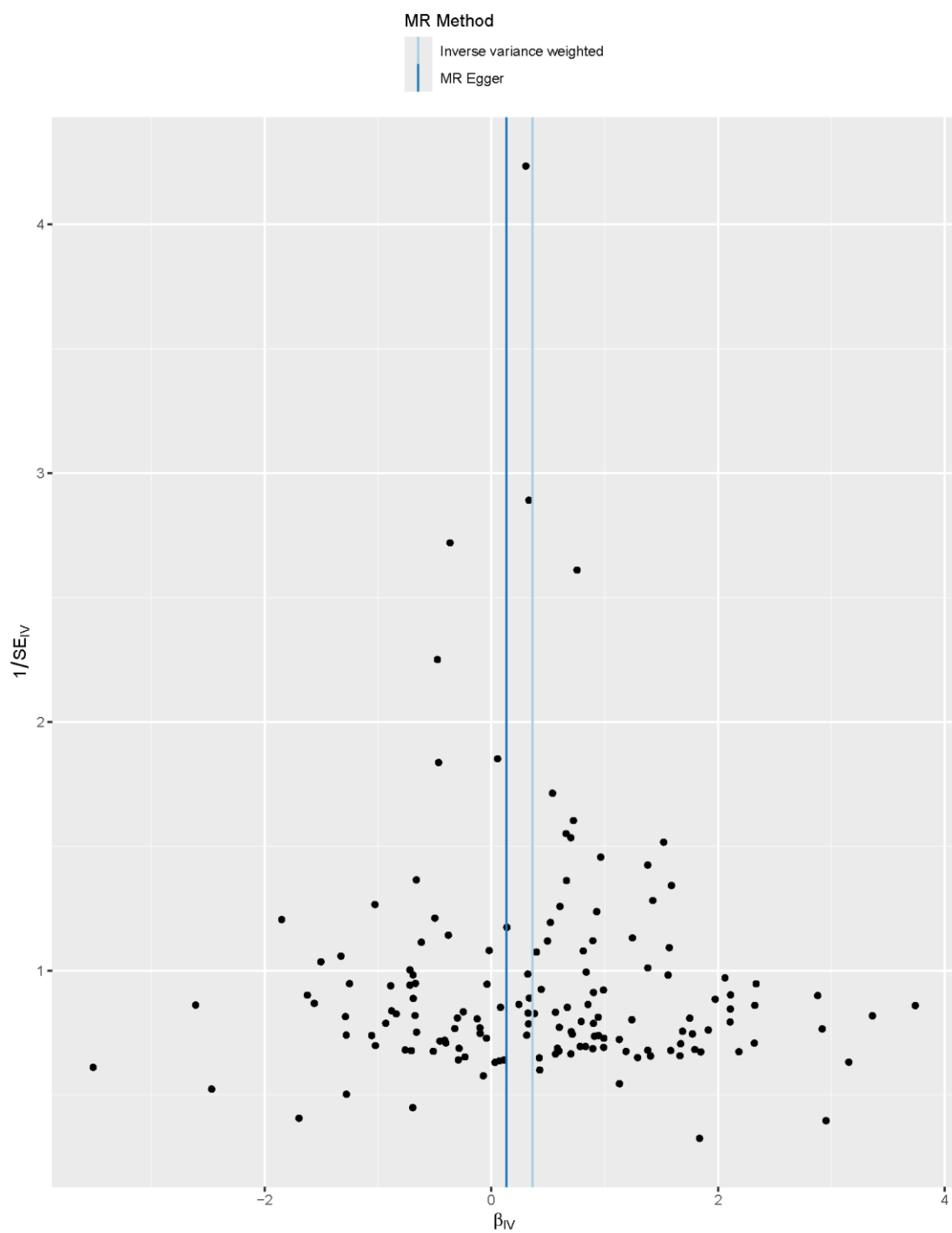


Figure S14. Funnel plots to visualize overall heterogeneity of MR estimates for the effect of serum uric acid on chronic kidney disease. MR, Mendelian Randomization.

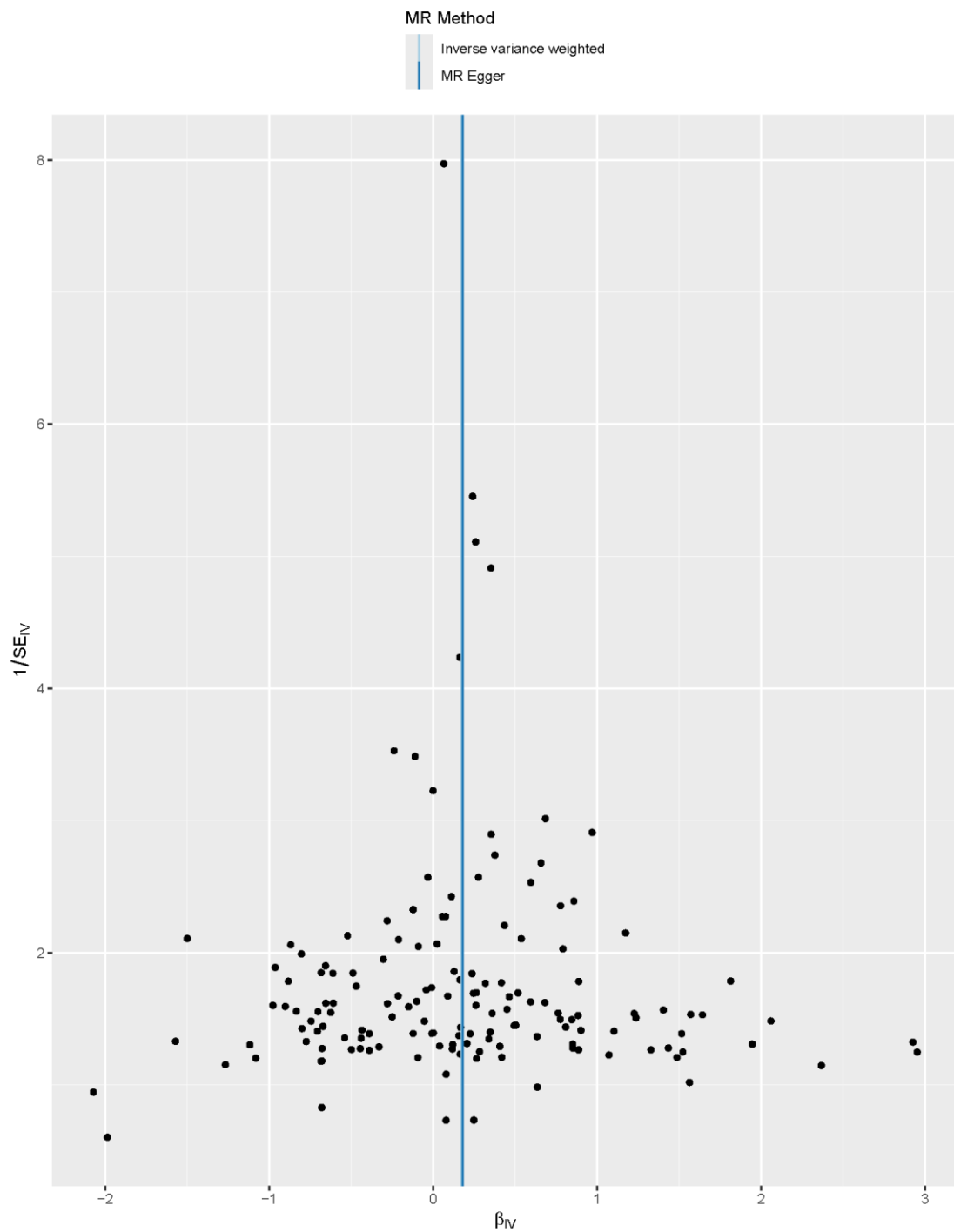


Figure S15. Funnel plots to visualize overall heterogeneity of MR estimates for the effect of serum uric acid on diabetes mellitus. MR, Mendelian Randomization.